

```
In [11]: import requests
from bot import BeautifulSoup
import pandas as pd
from bs4 import BeautifulSoup
import matplotlib.pyplot as plt
from sklearn.feature_extraction.text import CountVecorizer
from sklearn.decomposition import LatentDirichletAllocation
import seaborn as sns
import re
import nltk
from nltk.corpus import stopwords
from nltk.util import ngrams
from collections import Counter

In [12]: base_url = "https://www.airlinequality.com/airline-reviews/criticab/airways"
pages = 10
page_size = 100

all_reviews = []
all_data_dicts = []

headers = {
    "User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/87.0.4280.88 Safari/537.36"
}

for i in range(1, pages + 1):
    print(f"Scraping page {i}")

    url = f"{base_url}/page/{i}/tootby-post_data33DescpageSize={page_size}"

    response = requests.get(url, headers=headers)

    if response.status_code != 200:
        print(f"Failed to retrieve page {i}: Status code {response.status_code}")
        continue

    # Parse the HTML content
    parsed_content = BeautifulSoup(response.content, "html.parser")

    # Parse the reviews
    for para in parsed_content.find_all("div", {"class": "text-content"}):
        all_reviews.append(para.get_text().strip())

    # Parse the review stats
    for review_stats in parsed_content.find_all("div", {"class": "review-stats"}):
        rating = 1
        category = []

        for value in review_stats.find_all("td", {"class": "review-value"}):
            rating.append(value.get_text().strip())

        recommended = rating[-1]
        rating = rating[-1]

        for stars in review_stats.find_all("span", {"class": "review-rating-stars stars"}):
            star_count = len(stars.find_all("span", {"class": "star fill"}))
            rating.append(star_count)

        rating.append(recommended)

        for header in review_stats.find_all("td", {"class": "review-rating-header"}):
            category.append(header.get_text().strip())

        data_dict = dict(zip(category, rating))
        all_data_dicts.append(data_dict)

    print(f"----> {len(all_reviews)} total reviews")

df = pd.DataFrame(all_data_dicts)
df['reviews'] = all_reviews

print(df.head())

Scraping page 1
----> 100 total reviews
Scraping page 2
----> 200 total reviews
Scraping page 3
----> 300 total reviews
Scraping page 4
----> 400 total reviews
Scraping page 5
----> 500 total reviews
Scraping page 6
----> 600 total reviews
Scraping page 7
----> 700 total reviews
Scraping page 8
----> 800 total reviews
Scraping page 9
----> 900 total reviews
Scraping page 10
----> 1000 total reviews

Aircraft Type of Traveller      Seat Type \
0 Boeing 777-300ER              Business First Class
1 Boeing 777-300ER              Business Business Class
2 Boeing 787-9                  Solo Leisure Business Class
3 NAH                           Business Business Premium Economy
4 NAH                           Business Business Premium Economy

Route      Date Flown      Seat Comfort \
0 London to Seattle        April 2023      5.0
1 London Gatwick to Fortu...  March 2023      4.0
2 Cape Town to London      March 2023      5.0
3 London to Geneva         December 2024   1.0
4 London to New York       January 2025    1.0

Cabin Staff Service Food & Beverages Inflight Entertainment \
0 5.0 5.0 5.0
1 5.0 5.0 NAH
2 5.0 5.0 5.0
3 1.0 1.0 NAH
4 1.0 1.0 NAH

Ground Service Wifi & Connectivity Value For Money Recommended \
0 4.0 5.0 4 4 yes
1 4.0 NAH 4 yes
2 4.0 NAH 4 yes
3 1.0 NAH 1 no
4 5.0 NAH 1 no

reviews
0 Not Verified | LHR to BA Gold Wing worked well...
1 Not Verified | Very good service on this route...
2 Trip Verified | Flight mainly let down by a di...
3 Trip Verified | Another awful experience B...
4 Trip Verified | The service was rude, full...

In [13]: df.to_csv("BA5_Reviews.csv", index=False)

In [17]: remove_text = ["Trip Verified", "Not Verified"]
df['reviews'] = df['reviews'].replace(remove_text, '', regex=True)

In [19]: df.sample(15)

Out[19]:
Aircraft  Type of Traveller  Seat Type      Route      Date Flown      Seat Comfort  Cabin Staff Service  Food & Beverages  Inflight Entertainment  Ground Service  Wifi & Connectivity  Value For Money  Recommended  reviews
471      NAH      Solo Leisure      Premium Economy      London Heathrow to Athens Greece      April 2022      3.0      4.0      1.0      1.0      3.0      1.0      1      no      I find BA incredibly tacky and constantly l...
319      NAH      Couple Leisure      Economy Class      London City Airport to Frankfurt      July 2023      3.0      3.0      NaN      NaN      1.0      NaN      1      no      I A simple story with an unfortunate outcome ...
573      NAH      Couple Leisure      Premium Economy      London to Nairobi      June 2022      NaN      NaN      NaN      NaN      NaN      NaN      1      no      I booked a flight for myself and my wife fr...
503      Boeing 787-9 / A320      Solo Leisure      Economy Class      Montreal to Edinburgh via London Heathrow      January 2022      4.0      4.0      4.0      4.0      4.0      NaN      4      yes      I was pleasantly surprised that the airt...
159      A320      Solo Leisure      Business Class      London Heathrow to Brussels      March 2024      2.0      4.0      3.0      NaN      3.0      NaN      3      yes      I Boarding was difficult caused by west imp...
66      A321      Couple Leisure      Economy Class      Heathrow to Berlin      July 2024      3.0      4.0      3.0      NaN      4.0      NaN      3      yes      I Really terrible user experience. British A...
626      Boeing 787-9      Solo Leisure      Business Class      Chicago to London      March 2022      5.0      5.0      5.0      4.0      3.0      NaN      4      yes      I Flying LHR to CPT November 2022: BA ap...
128      NAH      Solo Leisure      Economy Class      Boston to London      May 2024      NaN      NaN      NaN      NaN      1.0      NaN      3      no      I Really terrible user experience. British A...
507      Boeing 777-300ER      Solo Leisure      Business Class      London to Cape Town      November 2022      4.0      2.0      1.0      4.0      1.0      NaN      1      no      I Flying LHR to CPT November 2022: BA ap...
169      NAH      Family Leisure      Economy Class      Washington to London      July 2022      1.0      1.0      NaN      NaN      1.0      NaN      1      no      I Washington to London. Our flight was unexpe...
319      A350      Solo Leisure      Economy Class      Austin to London Heathrow      April 2024      3.0      2.0      1.0      NaN      1.0      1.0      1      no      I Absolutely terrible. Lost my luggage on bo...
921      Boeing 777-300ER      Family Leisure      Economy Class      London to Mauritius      November 2019      1.0      4.0      1.0      1.0      1.0      1.0      2      no      I London to Mauritius. We are fortunate enoug...
781      NAH      Solo Leisure      Economy Class      London to Paris      September 2020      5.0      5.0      NaN      NaN      5.0      NaN      5      yes      I I entered the plane with a bunch of white f...
862      Boeing 777-300ER      Couple Leisure      Business Class      Singapore to Paris via London      January 2020      3.0      4.0      4.0      4.0      1.0      NaN      2      no      I Singapore to Paris via London. The flight...
446      A380      Business Class      Business Class      London to Dubai      February 2023      2.0      3.0      2.0      3.0      4.0      NaN      3      yes      I Boarding was decently organised. The A380 ...

In [11]: nltk.download('stopwords')

def preprocess_text(text):
    text = text.lower() # Convert to lowercase
    text = re.sub('[^a-zA-Z]', ' ', text) # Remove punctuation and special characters
    words = text.split() # Tokenize into words
    words = [word for word in words if word not in stopwords.words('english')] # Remove stopwords
    return words

df['clean_reviews'] = df['reviews'].apply(preprocess_text)

def generate_ngrams(texts, n=2): # Change n=2 for bigrams, n=3 for trigrams
    ngram_list = []
    for text in texts:
        ngram_list.extend(list(ngrams(text, n))) # Generate n-grams
    return ngram_list

n = 2 # Change to 3 for trigrams
ngrams_list = generate_ngrams(df['clean_reviews'], n=n)

ngram_freq = Counter(ngrams_list)

ngram_dict = {k: ' '.join(v) for k, v in ngram_freq.items()}

wc = WordCloud(width=1000, height=450, background_color='white').generate_from_frequencies(ngram_dict)

plt.figure(figsize=(15, 8))
plt.imshow(wc, interpolation='bilinear')
plt.axis('off')
plt.show()

In [14]: nltk_data Downloading package stopwords to
nltk_data C:\Users\hoboe\AppData\Local\NltkData\...
nltk_data Unzipping corpora\stopwords.zip.

flight london terminal 5 flight entertainment flight crew
connecting flight first time good flight learning gate
ba start though flight cancelled ba flight
ground never customer service
two hours return flight premium economy
club europe low cost 30 minutes flight attendant club world
flight ba food drink 3 hot 5 food drink another airline
british airways
london heathrow first class
via london cancelled flight next flight 1 hour
business class
flight attendants middle seat bottle water ba customer

In [15]: custom_colors = ("ff9999", "#6633ff", "#99ff99", "#ffcc99", "#ffcc33")

sco = df.groupby("Recommended")["reviews"].count().reset_index()
plt.pie(sco["reviews"], labels=sco["Recommended"], autopct='%1.1f%%',
        color=custom_colors, wedgeprops={'edgecolor': 'k'})

plt.legend()
plt.show()

no
68.70%
yes
31.30%

In [15]: df_sco = df[df["Recommended"] == "no"]
df_sco.describe()

Out[15]:
Seat Comfort  Cabin Staff Service  Food & Beverages  Inflight Entertainment  Ground Service  Wifi & Connectivity  Value For Money
count  601.000000      591.000000      485.000000      338.000000      632.000000      217.000000      687.000000
mean      2.096506      2.385787      1.843299      2.189349      1.753165      1.548387      1.505095
std      1.116843      1.317445      1.062208      1.219720      1.147828      0.980696      0.818635
min      1.000000      1.000000      1.000000      1.000000      1.000000      1.000000      1.000000
25%      1.000000      1.000000      1.000000      1.000000      1.000000      1.000000      1.000000
50%      2.000000      2.000000      1.000000      2.000000      1.000000      1.000000      1.000000
75%      3.000000      3.000000      3.000000      3.000000      2.000000      2.000000      2.000000
max      5.000000      5.000000      5.000000      5.000000      5.000000      5.000000      5.000000

In [17]: data = df_sco[df_sco["reviews"].str.contains("food | drink")]
data["reviews"] = data["reviews"].str.strip()

In [17]: review_word = []

for i in data["reviews"]:
    a = i.split(' ')
    for word in a:
        review_word.append(word)

unique_string = ' '.join(review_word)

wc = WordCloud(width=1000, height=450, background_color='white').generate(unique_string)

plt.figure(figsize=(15, 8))
plt.imshow(wc)
plt.axis('off')
plt.show()

aircraft ticket return told booked full dinner one minute
passenger available way enough extra food
option terminal british airways
good long business class people
gate even first class offered breakfast said water due half given really
hotel work mca al busines breakfast offered breakfast said water due half given really
offer mca al busines breakfast offered breakfast said water due half given really
cancel route better flying served late got already
paid found year went choice still board screen small first time
money say year went choice still board screen small first time
toilet grew poor cost now fly customer screen small first time
travel grew poor cost now fly customer screen small first time
last left flight attendant glass waiting wait nice came bag boarding every
eventually cancelled paid found year went choice still board screen small first time
eventually cancelled paid found year went choice still board screen small first time

In [21]: data = df_sco[df_sco["reviews"].str.contains("money | refund | value")]
data["reviews"] = data["reviews"].str.strip()

In [23]: review_word = []

for i in data["reviews"]:
    a = i.split(' ')
    for word in a:
        review_word.append(word)

unique_string = ' '.join(review_word)

wc = WordCloud(width=1000, height=450, background_color='white').generate(unique_string)

plt.figure(figsize=(15, 8))
plt.imshow(wc)
plt.axis('off')
plt.show()

another money
anything called
around go booking voucher one
wait charge london change
staff "connection" british airways
economy made moving phone offered extra
least changed money phone offered extra
customer week told u will
experience new service booked
even email seat cabin time airline
plane game rebook thing

In [25]: from nltk.tokenize import TextBlob
df['sentiment'] = df['reviews'].apply(lambda x: TextBlob(x).sentiment.polarity)
df['sentiment_label'] = df['sentiment'].apply(lambda x: 'positive' if x > 0 else 'negative' if x < 0 else 'neutral')

In [27]: print(df[['reviews', 'sentiment', 'sentiment_label']].head())

reviews sentiment \
0 LHR to BA Gold Wing worked well. Pleasant c... 0.402794
1 Very good service on this route BA2710 30th... -0.126488
2 Flight mainly let down by a disagreeable ... 0.207461
3 Another awful experience by British Airwea... -0.202050
4 The service was rude, full of attitude to... -0.045933

sentiment_label
0 positive
1 negative
2 positive
3 negative
4 negative

In [29]: df['sentiment_label'].value_counts().plot(kind='bar', color='skyblue')

plt.title('Sentiment Distribution')
plt.xlabel('Topic')
plt.ylabel('Number of Reviews')
plt.show()

Sentiment Distribution

Number of Reviews
600
500
400
300
200
100
0
positive negative neutral

In [31]: vectorizer = CountVecorizer(max_df=0.95, min_df=2, stop_words='english')
X = vectorizer.fit_transform(df['reviews'])

In [31]: lda = LatentDirichletAllocation(n_components=5, random_state=0)
lda.fit(X)

Out[31]:
v LatentDirichletAllocation
LatentDirichletAllocation(n_components=5, random_state=0)

In [33]: num_words = 16
for index, topic in enumerate(lda.components_):
    print(f"Topic {index+1}:")
    print([word for word, count in sorted(zip(topic.get_feature_names_out(), topic.get_feature_names_out()), key=lambda x: x[1], reverse=True)])

Topic #1:
['airline', 'flight', 'economy', 'experience', 'london', 'cabin', 'crew', 'comfort', 'food', 'seats', 'good', 'seat', 'time', 'staff', 'service', 'ba', 'flight']
Topic #2:
['help', 'delay', 'time', 'airline', 'booked', 'airport', 'cancelled', 'london', 'hours', 'cold', 'airways', 'customer', 'service', 'british', 'ba', 'flight']
Topic #3:
['business', 'passengers', 'lhr', 'ba', 'gate', 'chance', 'food', 'plane', 'economy', 'london', 'british', 'airways', 'passa', 'crew', 'seats', 'flight']
Topic #4:
['help', 'british', '30', 'late', 'hour', 'delayed', 'delay', 'ground', 'crew', 'plane', 'gate', 'heathrow', 'ba', 'luggage', 'flight', 'minutes']
Topic #5:
['passengers', 'lounge', 'boarding', 'club', 'value', 'seats', 'time', 'cabin', 'seat', 'service', 'good', 'food', 'crew', 'business', 'class', 'ba', 'flight']

In [37]: sentiment_counts = df['sentiment_label'].value_counts()
plt.pie(sentiment_counts, labels=sentiment_counts.index, autopct='%1.1f%%', startangle=140)
plt.title('Sentiment Distribution')
plt.show()

Sentiment Distribution

negative
37.5%
neutral
12%
positive
61.3%

In [39]: topic_counts = lda.transform(X).argmax(axis=1)
plt.hist(topic_counts, bins=4, edgecolor='k')
plt.title('Topic Frequency')
plt.xlabel('Topic')
plt.ylabel('Number of Reviews')
plt.show()

Topic Frequency

Number of Reviews
400
350
300
250
200
150
100
50
0
0.0 0.5 1.0 1.5 2.0 2.5 3.0 3.5 4.0
Topic

In [41]: print(df[['sentiment_label'] == 'positive'].head())
print(df[['sentiment_label'] == 'negative'].head())

Aircraft Type of Traveller      Seat Type      Route \
0 Boeing 777-300ER              Business First Class      London to Seattle
1 Boeing 787-9                  Solo Leisure Business Class      Cape Town to London
2 Boeing 777-300ER              Business Premium Economy      Gatwick to Antigua
3 NAH                           Couple Leisure Premium Economy      Mumbai to London
4 A350                          Couple Leisure Business Class      Nairobi to London

Date Flown      Seat Comfort  Cabin Staff Service  Food & Beverages \
0 April 2023      5.0      5.0      5.0
1 March 2023      5.0      2.0      5.0
2 March 2023      5.0      5.0      4.0
3 February 2023      5.0      4.0      3.0
4 February 2023      4.0      1.0      2.0

Inflight Entertainment Ground Service Wifi & Connectivity \
0 5.0 5.0 4.0 4.0 3.0
1 5.0 5.0 4.0 4.0 1.0
2 4.0 5.0 5.0 4.0 4.0
3 5.0 5.0 5.0 4.0 3.0
4 3.0 4.0 4.0 NAH

Value For Money Recommended \
0 4 yes
1 4 yes
2 5 yes
3 5 yes
4 3 no

reviews
0 LHR to BA Gold Wing worked well. Pleasant c...
1 Flight mainly let down by a disagreeable ...
2 This time British Airways managed to get ...
3 The seats were excellent, with a feel of ...
4 Prior to boarding a gate agent seemed to ...

clean_reviews sentiment \
0 lhr, ts, ba, gold, wing, worked, well, please... 0.402794
1 flight mainly let down by a disagreeable... -0.126488
2 this time british airways managed to get... 0.207461
3 seats excellent feel much room official... 0.370966
4 nightmare getting through security nairobi... 0.051718

sentiment_label
0 positive
1 positive
2 positive
3 positive
4 negative

Aircraft Type of Traveller      Seat Type \
0 A320 Couple Leisure Business Class
1 NAH Solo Leisure Business Class
2 NAH Business Premium Economy
3 NAH Business Business Class
4 A320 Solo Leisure Economy Class

Route      Date Flown      Seat Comfort \
1 London Gatwick to Fortu... March 2023      4.0
2 London to Geneva December 2024      1.0
3 London to New York January 2025      1.0
4 Chesham to London January 2025      1.0
5 London Gatwick to Marrakech February 2025      4.0

Cabin Staff Service Food & Beverages Inflight Entertainment \
1 5.0 5.0 NAH
2 1.0 1.0 NAH
3 1.0 1.0 NAH
4 1.0 1.0 NAH

Ground Service Wifi & Connectivity Value For Money Recommended \
1 4.0 4 NAH 4 yes
2 1.0 1 NAH 1 no
3 1.0 1 NAH 1 no
4 2.0 2 NAH 2 no

reviews
1 Very good service on this route BA2710 30th...
2 Another awful experience by British Airwea...
4 The service was rude, full of attitude to...
5 This flight was a joke. There was four pe...
6 Prior to boarding a gate agent seemed to ...

clean_reviews sentiment \
1 good service, route, ba2710, 30th, march, ea... -0.126488
2 another awful experience by british airways... -0.202050
4 service, rude, full, attitude, food, poorly... -0.045933
5 flight, joke, four, people, business, class... -0.458033
6 prior, boarding, gate, agent, seemed, sick... -0.076786

sentiment_label
1 negative
2 negative
4 negative
5 negative
6 negative

In [43]:
```